AP PRECALCULUS

Unit 1 • Worksheet 1.2C

Worksheet 1.2C • Rates of Change

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$V(t)$ is the volume of water in a tank, in liters, t minutes after a valve opens. What are the units of the average rate of change of V?

(A) minutes per liter

(B) liters

(C) liters per minute

(D) liters times minutes

Question 2

NO CALCULATOR

The population of a town was 12,400 in 2010 and 15,100 in 2019. Which is the average rate of change of the population over this period?

(A) 270 people per year

(B) 300 people per year

(C) 2700 people per year

(D) 337.5 people per year

Question 3

NO CALCULATOR

A car's distance from home is $D(t)$ km, t hours after noon. $D(1) = 80$ and $D(3) = 20$. Which is the best interpretation of $\frac{D(3)-D(1)}{3-1}$?

(A) The car traveled 30 km.

(B) Between 1 p.m. and 3 p.m., the car's distance from home decreased on average by 30 km per hour.

(C) At 3 p.m. the car was moving at 30 km per hour.

(D) The car was 30 km from home at 2 p.m.

Question 4 (Free response) NO CALCULATOR

The temperature T (in °F) in a city is recorded at times t (hours after midnight).

t	0	4	8	12	16
$T(t)$	52	48	57	71	66

(a) Find the average rate of change of T on $[4, 12]$. Interpret it in context with units.
(b) On which interval of length 4 did the temperature increase the fastest on average?
(c) Find $\frac{T(16)-T(0)}{16-0}$ and explain why it does not describe how the temperature changed during the day.